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PROFESSIONAL EXPERIENCES

Department of Atmospheric Sciences, University of Hawai'i at Mānoa Postdoctoral Researcher	Jan. 2026–Present
Department of Atmospheric Sciences, Yonsei University Postdoctoral Researcher	Mar. 2025–Dec. 2025

RESEARCH INTERESTS

- Characteristics of turbulence in the free atmosphere estimated using observation data
- Sources, generation mechanisms, and climatological variability of atmospheric turbulence
- Development and validation of observation-based turbulence estimation methods
- Aviation turbulence forecasting, evaluation, and model improvement using aircraft and radiosonde observations
- Impacts of climate change and solar geoengineering on atmospheric turbulence and aviation hazards

EDUCATION

Integrated Master's and Ph.D. Yonsei University, Seoul, South Korea Dissertation title: Characteristics and Sources of Turbulence in the Free Atmosphere and its Application to Aviation Turbulence (Advisor: Professor Hye-Yeong Chun)	Sep. 2017–Feb. 2025
BSc in Atmospheric Sciences Yonsei University, Seoul, South Korea	Mar. 2012–Aug. 2017

PUBLICATIONS

- **Ko, H.-C.**, and H. Sun (Major revision). Global variability and recent increase in stratospheric turbulence revealed by radiosonde observations.
- **Ko, H.-C.**, and H.-Y. Chun (*Accepted in **Geophysical Research Letters***). A generation mechanism of Thorpe-detected turbulence: A case study using WRF model simulation.

- **Ko, H.-C.**, and H.-Y. Chun (Accepted in *npj Climate and Atmospheric Science*). Increasing trend of vertical wind shear revealed by high-resolution radiosonde data over the United States, <https://doi.org/10.1038/s41612-026-01466-w>.
- **Ko, H.-C.**, H.-Y. Chun, and H.-K. Lee (2026). Can climate geoengineering mitigate the rising threat of aviation turbulence? *Environmental Research Letters*, 21(12), 124020, <https://doi.org/10.1088/1748-9326/ae7b5c>.
- Jung, H.-W., H.-Y. Chun, and **H.-C. Ko** (2026, in Korean). Generation Mechanisms of Two Aviation Turbulence Events Leading to Aircraft Accidents in Korea. *Atmosphere-Korea*, 36(2), 91–115, <https://doi.org/10.14191/Atmos.2026.36.2.091>.
- **Ko, H.-C.**, and H.-Y. Chun (2026). A New Method for Estimating Atmospheric Turbulence from Global High-Resolution Radiosonde Data and Comparison with the Thorpe Method. *Earth System Science Data*, 18(3), 1905–1920, <https://doi.org/10.5194/essd-18-1905-2026>.
- **Ko, H.-C.**, H.-Y. Chun, and P. Bechtold (2025). Evaluation and improvement of the ECMWF aviation turbulence forecasts. *Journal of Geophysical Research: Atmospheres*, 130(16), e2024JD043158, <https://doi.org/10.1029/2024JD043158>.
- **Ko, H.-C.**, H.-Y. Chun, M. A. Geller, and B. Ingleby (2024). Global distributions of atmospheric turbulence estimated using operational high vertical-resolution radiosonde data. *Bulletin of the American Meteorological Society*, 105(12), E2551–E2566, <https://doi.org/10.1175/BAMS-D-23-0193.1>.
- Lee, Y.-S., H.-Y. Chun, and **H.-C. Ko** (2023). Lower tropospheric states revealed in high vertical-resolution radiosonde data in Korea and synoptic patterns for instability based on a self-organizing map. *Atmospheric Research*, 295, 107037, <https://doi.org/10.1016/j.atmosres.2023.107037>.
- **Ko, H.-C.**, H.-Y. Chun, R. D. Sharman, and J.-H. Kim (2023). Comparison of eddy dissipation rate estimated from operational radiosonde and commercial aircraft observations in the United States. *Journal of Geophysical Research: Atmospheres*, 128, e2023JD039352, <https://doi.org/10.1029/2023JD039352>.
- **Ko, H.-C.**, and H.-Y. Chun (2022). Potential sources of atmospheric turbulence estimated using the Thorpe method and operational radiosonde data in the United States. *Atmospheric Research*, 265, 105891, <https://doi.org/10.1016/j.atmosres.2021.105891>.
- **Ko, H.-C.**, H.-Y. Chun, R. Wilson, and M. A. Geller (2019). Characteristics of atmospheric turbulence retrieved from high vertical-resolution radiosonde data in the United States. *Journal of Geophysical Research: Atmospheres*, 124, 7553–7579, <https://doi.org/10.1029/2019JD030287>.

RESEARCH HIGHLIGHTS AND MEDIA COVERAGE

- **Ko** et al. (2026, *Environmental Research Letters*) featured in Eos: “Stratospheric Aerosol Injection Might Make Flights Smoother” (22 July 2026), [link](#).
- **Ko** et al. (2026, *Earth System Science Data*) featured in the Yonsei University Research Gallery (23 March 2026), [link](#).
- **Ko** et al. (2019, *Journal of Geophysical Research: Atmospheres*) selected for an Eos Editor’s Highlight (16 August 2019), [link](#).

PROFESSIONAL SERVICES

Journal reviewer for

- *Atmospheric Chemistry and Physics*
- *Journal of Aircraft*
- *Meteorology and Atmospheric Physics*
- *Journal of Geophysical Research: Atmospheres*
- *Journal of Climate*
- *Geophysical Research Letters*
- *Atmospheric Research*
- *Asia-Pacific Journal of Atmospheric Sciences*

VISITING RESEARCH COLLABORATION

- **European Centre for Medium-Range Weather Forecasts (ECMWF)**, invited by Dr. Peter Bechtold Mar. 1–May 31, 2024
- **National Center for Atmospheric Research (NCAR)**, hosted by Dr. Robert D. Sharman Jan. 16–Mar. 15, 2023

INTERNATIONAL CONFERENCE PRESENTATIONS

- **Ko, H.-C.**, H.-Y. Chun, and P. Bechtold (2025). Evaluation and Improvement of the ECMWF Aviation Turbulence Forecasts. **2025 APARC GWs and FISAPS Symposium**, Seoul, South Korea. [Poster]
- **Ko, H.-C.**, and H.-Y. Chun (2025). A New Estimation of Atmospheric Turbulence Using Global High Vertical-Resolution Radiosonde Data. **2025 APARC GWs and FISAPS Symposium**, Seoul, South Korea. [Oral]
- **Ko, H.-C.**, H.-Y. Chun, M. A. Geller, and B. Ingleby (2024). Atmospheric Turbulence Retrieved from Global High Vertical-Resolution Radiosonde Data. **21st Annual Meeting of AOGS**, Pyeongchang, South Korea. [Oral]
- **Ko, H.-C.**, H.-Y. Chun, R. D. Sharman, and J.-H. Kim (2023). Estimation of Turbulence Using Operational High Vertical-Resolution Radiosonde Data and Comparison with Aircraft Turbulence Reports, **IWAWF 2023**, Gyeongju, South Korea. [Oral]
- **Ko, H.-C.**, H.-Y. Chun, R. D. Sharman, and J.-H. Kim (2023). Estimation of Turbulence Using Operational HVRD and Comparison with Aircraft Turbulence Reports. **FISAPS**, Boulder, Colorado, USA. [Oral]
- **Ko, H.-C.**, H.-Y. Chun, R. D. Sharman, and J.-H. Kim (2023). Atmospheric turbulence estimated from high-resolution operational radiosonde data and comparison with in-situ flight observations. **20th Annual Meeting of AOGS**, Suntec, Singapore. [Oral]
- **Ko, H.-C.**, H.-Y. Chun, R. D. Sharman, and J.-H. Kim (2023). Estimation of Aviation Turbulence Using Operational High Vertical-Resolution Radiosonde Data and Comparison with In-Situ Flight EDR. **103rd AMS annual meeting 2023**, Denver, Colorado, USA. [Oral]
- **Ko, H.-C.**, and H.-Y. Chun (2022). Estimation of Eddy Dissipation Rate (EDR) Using Radiosonde Data and Comparison with In-Situ Flight EDR. **19th Annual Meeting of AOGS**, online. [Oral]
- **Ko, H.-C.**, and H.-Y. Chun (2021). Atmospheric Turbulence Estimated Using Thorpe Method and Operational Radiosonde Data in USA and Its Potential Sources. **18th Annual Meeting of AOGS**, online. [Oral]

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- **Ko, H.-C.**, and H.-Y. Chun (2021). Atmospheric Turbulence Estimated using Thorpe Method and Operational Radiosonde Data in USA and Its Potential Sources, **101st AMS annual meeting 2021**, online. [Oral]
 - **Ko, H.-C.**, H.-Y. Chun, and R. D. Sharman (2019). Characteristics of atmospheric turbulence retrieved from high vertical-resolution radiosonde data in the United States and comparison with observed aviation turbulence, **16th Annual Meeting of AOGS**, Suntec, Singapore. [Oral by Prof. Chun]
 - **Ko, H.-C.**, H.-Y. Chun, M. A. Geller, and P. T. Love (2018). Characteristics of turbulence in the free atmosphere retrieved from high vertical-resolution radiosonde data in US. **FISAPS**, Kühlungsborn, Germany. [Oral]
 - **Ko, H.-C.**, H.-Y. Chun, M. A. Geller, and P. T. Love (2017). Atmospheric turbulence retrieved from high vertical-resolution radiosonde data in USA and comparison with in-situ flight data. **FISAPS, QBOi, and SATIO-TCS Joint Workshop**, Kyoto, Japan. [Poster]

GRANTS AND AWARDS

Excellent Thesis Award Yonsei University	2026
Best Graduate Student Paper Award (Sung-Ki Chung Alumni Award) Yonsei University	2025
Outstanding Dissertation Award Korean Meteorological Society	2025
The Best Student Paper and Presenter American Meteorological Society	2023
Outstanding Academic Paper Award Yonsei University	2022
Research Subsidies for Ph.D. Candidates National Research Foundation of Korea	Jun. 2020– May 2022
Outstanding Academic Paper Award BK21Plus Institute of Earth-Atmosphere-Astronomy	2020
Outstanding Academic Paper Award Yonsei University	2019
Ko et al. (2019, JGR-A) is selected on Eos Editor's Highlights Eos	2019

PROJECT

Characteristics and sources of atmospheric turbulence estimated using high-resolution operational radiosonde data funded by National Research Foundation of Korea	Jun. 2020– May 2022
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TECHNICAL SKILLS

- Data processing and Visualizing: Fortran and NCL
- OS: Linux/Unix
- Numerical model: WRF, ECMWF IFS

REFERENCES

Professor Hye-Yeong Chun

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Dr. Peter Bechtold

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